Q1 - Stereo hearing. Stereo systems often use a single subwoofer that plays low-frequency sounds. The reasoning is that we don't hear the direction of bass sounds, so there's no point using two subwoofers. Below what frequency will we struggle to tell the direction of a sound? Assume the human head is 25cm across and our brain can detect when two signals are a radian out of phase.

Q2 - More localization. We've talked about time delays for localizing sounds, but our brain uses several other clues, including loudness. If a sound is on one side of our head then our head blocks the sound from the ear on the far side from the sound. You might think this would let you localize bass, but in fact you still can't. Please explain why, and estimate the frequency below which this won't work. You may assume your head has not changed size since the previous problem.

Q3 - Two cardinals are sitting in a tree 20m away. The cardinals are 0.5m apart. What is the angle separating the cardinals, assuming the branch they're sitting in is perpendicular to your line of sight?
One of the cardinals starts singing. Assume their song has lots of notes that go up to 3.4 kHz. Without looking, could you tell which one is singing?

Q4 - Modes of a room. You find yourself in a square room that is 6.8m on a side and 3.4m tall.
A) What are the fundamental frequencies of the room along the principle directions? (i.e. length/width (which will be the same) and height)
B) Remember that the frequencies of a rectangular room are $\sqrt{m_x^2\nu_x^2 + m_y^2\nu_y^2 + m_z^2\nu_z^2}$ where ν_x, ν_y, ν_z are the fundamental frequencies in each direction and m_x, m_y, m_z are arbitrary non-negative integers. If we want to find all resonant frequencies of the room *below* 100 Hz (so not including 100 Hz), what values can each of m_x, m_y, and m_z have?
C) For those who feel like a challenge, how many *distinct* resonant frequencies will our room have below 100 Hz, and what are the corresponding values of m_x, m_y, and m_z?

Q5 - Mode spacing on a string. Assume you have a string clamped at both ends that has n harmonics below some cutoff frequency ν_{cut}. How many harmonics will there be below $2\nu_{cut}$?

Q6 - Mode spaing in a room. Redo problem 5, but for a roughly (but not exactly) cubical room. Hint - if m_x, m_y, and m_z are less than some value m_{cut}, then the total number of modes is roughly m_{cut}^3 because there are m_{cut} possibilities for each of m_x, m_y and m_z.

Discussion of Q5 and Q6: You should find that if we double the cutoff frequency on a string you get twice as many modes. That means the number of modes between 0 and ν_{cut} is the same as the number of modes between ν_{cut} and $2\nu_{cut}$. The average *separation* between the frequencies of the harmonics is the same. For the room, you should find there are eight times as many modes going to $2\nu_{cut}$, so the number of frequencies between ν_{cut} and $2\nu_{cut}$ is seven times as many as between 0 and ν_{cut}. The separation between mode frequencies gets smaller as we go to higher frequency, and eventually they will get so closely spaced that the room no longer acts like a resonator with distinct frequencies.